

ספר פרויקט – מאמן עסקי

חלק 1 – סקירה ספרותית

Idea: Startup Coach with LLM base on ChatGPT.

Name: נדב דניאל

Course Name: פרויקט גמר במדעי המחשב

Source 1: Book - ChatGPT Coach AI: Your AI-Powered Coaching Assistant

Contribution to the project: The book provides the business and operational justification for the project. It presents a practical and practitioner-proven framework for using ChatGPT as a personal coaching assistant. Its main contribution lies in demonstrating the need for a 24/7 available AI coach and its ability to handle routine tasks (such as formulating basic strategy and reflection), which allows the entrepreneur to focus on more complex problems.

Major subjects: The use of LLM models (including ChatGPT) as personal coaching tools for AI. Engineering prompts to create a professional business coach persona for AI through targeted prompts. Addressing access and cost issues associated with traditional human coaching through scale and availability.

Summary: The book presents a practical approach to adopting ChatGPT as an AI-Powered Coaching Assistant. It describes how coaches and entrepreneurs can leverage AI to break through their time and impact ceiling. The book focuses on Prompt Engineering techniques to transform the AI from a generic model into a mentor that assists in habit building, goal setting, and plan formulation. The main conclusion is that AI is an Assistant that organizes the chaos and frees up time for deep thinking, but it does not replace the emotional human touch. It emphasizes the need to maintain full Context for the artificial coach to be effective.

Gap between works: The book focuses on practical implementation (How-To) and does not present data-driven empirical research on performance comparison or the long-term effectiveness of the digital coach. It does not propose an advanced technological architecture like RAG (Retrieval-Augmented Generation) to deal with hallucination issues or the need for a custom knowledge base for businesses, thus providing theoretical support for the need for the engineering solution of the current project.

Bibliography list MLA Format: Juneja, Ashish. *ChatGPT Coach AI: Your AI-Powered Coaching Assistant*. 2024.

Source 2: URL - CoachVox AI

Name of article: "ChatGPT as a Business Coach: How to Prompt ChatGPT to Be Your Business Coach."

Contribution for the project: This source is a living example of an industrial application of the project concept, providing critical insights into Prompt Engineering and persona design. Its contribution lies in demonstrating the market demand for such solutions and detailing the

prompting techniques required to upgrade a generic model into a focused, goal-oriented coach. It reinforces the need to focus on persona and 24/7 availability.

Major subjects: The demand for LLM-based tools in the field of business coaching can be seen in the market demand for AI Coaching. Prompt Engineering for Specialization consists of the creation of professional coach personas (Role-Playing Prompts). Conversational Design is the process of designing a conversational interface (Chatbot) for focused business use.

Summary: The article presents a practical guide to creating a ChatGPT-based business coach. It explains that successful coaches require specific skills and approaches (goal setting, creating an action plan, reflective questions), and demonstrates how to embed these skills into the prompt. The emphasis is on using techniques like "System Prompt" to guide the model to maintain a consistent, professional tone and operate under a specific methodology. The source reinforces the idea that AI can become an effective and available supportive tool, but it is entirely dependent on the quality of the prompt provided to it.

Gap between works: As a practical and industrial source, it does not include academic methodology and empirical data. There is no mention of factual accuracy or outdated knowledge issues in Hallucination/Accuracy. The article does not provide solutions to my project's core engineering problems. Further, this article only discusses prompting and not the need to enhance specialization through integration of a broader, external knowledge base. This reinforces the need for a RAG architecture.

Bibliography list MLA Format:

"ChatGPT as a Business Coach: How to Prompt ChatGPT to Be Your Business Coach."

CoachVox AI, www.coachvox.ai/chatgpt-as-a-business-coach/. Accessed 27 Nov. 2025.

Source 3: Research Article

Name of article: The Effect of ChatGPT on Business Success

Contribution for the project: This is a critical empirical source that provides quantitative validation for the project's core hypothesis that the strategic use of ChatGPT positively impacts business success. The article identifies Customization and User Satisfaction as the main areas that must be prioritized. This finding is a direct justification for the project's need to build a RAG architecture or another mechanism that allows for the customization of a knowledge base for startups, thereby meeting the requirements for accuracy and satisfaction that the article identifies as key to success.

Major subjects: The major subjects that should be covered by the literature review based on this article are the Quantitative Impact of AI (quantitative evidence of ChatGPT's effect on business performance), Customization as a Success Factor (the importance of tailoring the AI tool to the business's needs), User Satisfaction Metrics (using user satisfaction metrics in accuracy, overall performance, and recommendation to evaluate AI tools), and the Strategic AI Adoption (the necessity of a clear strategy for integrating AI tools into a business).

Summary: The study conducted quantitative research (a survey) to investigate the role of ChatGPT in business success and the effectiveness of its adoption. The findings indicate that

companies need to formulate a strategy for using ChatGPT, and that user satisfaction must be prioritized. The article concludes that the customization of the AI is vital for promoting success, and there is a need for proactive, tool-driven decision-making. The research supports that strategic adoption of ChatGPT can enhance business operations, overall customer satisfaction, and total success in the rapidly changing business landscape.

Gap between works: This is an article in Business Management (Business Review) and not Computer Science. As such, it identifies the need for "Customization" and "Accuracy" as success factors, but it does not propose technological solutions for achieving them. It does not discuss specific architectures like Retrieval-Augmented Generation (RAG), Fine-Tuning, or the use of Vector Databases. The project precisely fills this gap by developing the engineering architecture required to realize the business requirements (Customization & Accuracy) identified in this research.

Bibliography list MLA Format:

Maalouf, Georges Y. "The Effect of ChatGPT on Business Success." Intern. Journal of Profess. Bus. Review, vol. 8, no. 12, 2023, pp. 01–19, e04134.

Source 4: Research Article

Name of article: Is ChatGPT Better at Business Consulting than an Expert? A Comparative Study.

Contribution for the project: This is a critical comparative source that provides the foundation for the Evaluation of my project business coach. The article provides academic validation that AI can compete with human consultants in defined consulting tasks. Its contribution is in the Justification of the Project, as it proves that AI can provide expert value in business, and in Defining Limitations, as it identifies when and where the AI still fails compared to a human (typically in complex, emotional, or creative tasks), thus defining the gap my project needs to overcome.

Major subjects: The major subjects that should be covered by the literature review based on this article include AI vs. Human Consulting (empirical comparison between AI performance and a human consultant), Evaluation of AI Advice Quality (metrics for assessing the quality of business advice from language models), and Limitations of LLMs in Complex Tasks (identifying the limits of AI in complex and long-term consulting tasks).

Summary: The study empirically compares the quality of advice provided by ChatGPT to that given by expert human business consultants. The findings showed that in specific, focused consulting tasks, the advice from ChatGPT was comparable to the quality of human advice, and in some cases even surpassed it. However, the article emphasizes that the AI struggles with tasks requiring emotional understanding, high creativity, or complex social contexts, where human experts are still superior. The conclusion is that AI is a useful tool but does not entirely replace the expert human touch.

Gap between works: The research focuses on short, one-off interactions and does not examine the effectiveness of AI in a Long-Term Coaching relationship, which is the focus of a yearly final project. Similarly, it examines a generic model (ChatGPT) and does not examine

the difference created when the AI is anchored to a dedicated knowledge base (like RAG). my project fills this gap by examining a Customized AI system (Customized RAG) designed to support prolonged processes.

Bibliography list MLA Format:

Jarco, Michał, et al. "Is ChatGPT Better at Business Consulting than an Expert? A Comparative Study." *Forum Scientiae Oeconomia*, vol. 11, no. 2, 2023, pp. 49–66.

Source 5: Research Article

Name of article: Examining the Applications of Intelligent Tutoring Systems in Real-World Educational Settings: A Meta-Analysis of 40 Studies.

Contribution for the project: This is a critical meta-analysis that forms the pedagogical/instructional basis of my project. Since my business coach is essentially a mentoring and learning system, I must prove that it is founded on proven educational principles. This source provides Justification for the Structure, as the meta-analysis proves that ITS systems consistently improve learning outcomes, which justifies my need to build the coach with a goal-oriented and adaptive structure (like ITS). Furthermore, since my project deals with Long-Term Coaching, ITS provides the research framework for understanding how to support continuous learning through AI.

Major subjects: The major subjects that should be covered by the literature review based on this article include Intelligent Tutoring Systems (ITS) (core principles of ITS such as adaptive feedback and goal-oriented learning), AI for Long-Term Learning (evidence for using AI to support continuous, non-one-off learning), and Educational Frameworks in AI Coaching (the argument that proven educational frameworks like ITS must be used to design the coaching/mentoring experience).

Summary: This article presents a comprehensive meta-analysis of 40 studies that examined applications of ITS in real-world educational settings. The central finding is that using ITS leads to a consistent and significant improvement in learning outcomes compared to traditional instructional methods. These systems are particularly effective due to their ability to provide Adaptive Feedback, adjust the difficulty level of tasks to the learner's needs, and track their progress over time. The meta-analysis emphasizes that the effectiveness of ITS largely depends on the quality of interaction and content design.

Gap between works: The research is in an Educational Setting and not in the business environment of entrepreneurship and startups. While it proves the structure's effectiveness, it does not address the unique complexities of entrepreneurship: high uncertainty, Unstructured Business Data, and Affective Load (emotional burden). My project fills this gap by taking the structural principles of ITS (adaptive and focused learning) and implementing them using a RAG architecture within the unique business context of startups.

Bibliography list MLA Format: Wang, Yu, et al. "Examining the Applications of Intelligent Tutoring Systems in Real-World Educational Settings: A Meta-Analysis of 40 Studies." *Education and Information Technologies*, vol. 28, 2023, pp. 10351–10386.

Source 6: Systematic Literature Review

Name of article: Artificial Intelligence-Empowered Conversational Agents in Business: A Systematic Literature Review.

Contribution for the project: This is an essential source that establishes my project within the field of Conversational AI and provides a systematic review of the benefits and challenges of AI-based conversational agents in a broad business context. Its contribution focuses on Defining the Value Proposition, as it identifies the areas where AI agents bring the highest business value (such as decision-making and operational efficiency), Adoption Drivers, as it provides a theoretical background for why entrepreneurs would adopt my business coach (availability, speed, and information handling capability), and Organizational Challenges, as it identifies the organizational and technological obstacles that must be considered (e.g., privacy, inability to handle emotion).

Major subjects: The major subjects that should be covered by the literature review based on this article include Conversational Agent Value (the business value of AI-driven conversational agents), Business Decision-Making Support (the role of conversational agents in supporting and improving decision-making processes), and Future Business Applications (future trends in the application of conversational AI in business).

Summary: The study provides a comprehensive systematic literature review on the application of Artificial Intelligence-empowered conversational agents (such as Chatbots) in business environments. The article concludes that conversational agents are game-changers in several fields, such as customer service, marketing, and improving operational efficiency. The main benefits are the ability to provide Personalization, 24/7 availability, and rapid data analysis. However, the review emphasizes that successful implementation requires addressing ethical and strategic challenges, and that the models still struggle with tasks requiring empathy and complex emotional understanding.

Gap between works: The review is generic and broad, referring to conversational agents across a wide range of industries (retail, healthcare, finance). It does not focus on or delve into the unique and dynamic needs of early-stage entrepreneurship and startups. Furthermore, the review does not discuss RAG architecture or specific engineering solutions for Knowledge Grounding that are required to transform the conversational agent into an expert and reliable coach.

Bibliography list MLA Format: Mariani, Marcello M., et al. "Artificial Intelligence-Empowered Conversational Agents in Business: A Systematic Literature Review." *Journal of Business Research*, vol. 161, 2023, article 113850.

Source 7: Research Article

Name of article: Conversational and Generative Artificial Intelligence and Human-Chatbot Interaction.

Contribution for the project: This source focuses directly on the quality of interaction between the user and the AI, which is a critical component for the success of my business coach (User Experience and Engagement). It provides Justification for the UX by offering insights into the necessary components to make the interaction natural, relational, and persuasive, supporting the project's UX/UI and Prompt Engineering requirements. Furthermore, it helps define the required Engagement Metrics in my Performance Metrics & Evaluation section, such as user retention and emotional engagement.

Major subjects: The major subjects that should be covered by the literature review based on this article include Human-Chatbot Interaction (HCI) (analysis of the interface and communication between a human and a conversational system), Generative AI in Conversations (improving conversational capability through Generative LLMs), and Trust and Adoption (factors influencing user trust and long-term adoption of conversational agents).

Summary: The article reviews the impact of Conversational AI and Generative AI on human-chatbot interactions. It highlights how LLMs have significantly improved the Fluency, Context, and overall capability of conversational agents. The main findings show that successful interactions rely on the AI's ability to maintain a continuous dialogue, understand complex intentions, and form a minimal connection with the user to enhance engagement. The article calls for developers to focus on conversational design that encourages trust and transparency regarding the AI's capabilities.

Gap between works: The article is excellent in covering the interaction aspect, but it is generic in terms of the problem domain and does not specifically address the challenges of customized knowledge or factual accuracy required in business consulting (where a small mistake could be critical for a startup). My project fills this gap by implementing the RAG architecture to ensure that a natural interaction (as recommended here) is also accurate and Grounded in reliable business data.

Bibliography list MLA Format: Akpan, Ikpe Justice, et al. "Conversational and Generative Artificial Intelligence and Human-Chatbot Interaction." *International Transactions in Operational Research*, vol. 31, no. [The volume/issue information is incomplete in the source provided].

Source 8: Research Article

Name of article: Creating Narrative Scripts Using ChatGPT to Strengthen Entrepreneurial Skills.

Contribution for the project: This article provides essential academic justification for the active skill development component of the AI Coach. It moves beyond simple advice and demonstrates empirically that generative AI can function as a creative collaborator to enhance vital entrepreneurial competencies, such as narrative development, pitching, and opportunity articulation. This supports the project's goal of improving actual founder performance and aligns with educational best practices for entrepreneurial training.

Major subjects: The major subjects that should be covered by the literature review based on this article include AI in Entrepreneurial Education (the use of AI tools to facilitate learning in entrepreneurship), Generative AI and Creativity (how LLMs assist in structured creative tasks like scripting and storytelling), and Entrepreneurial Skills Development (the direct impact of AI on core founder competencies like pitching and narrative construction).

Summary: The study investigates the impact of using ChatGPT to assist students in creating structured narrative scripts relevant to entrepreneurship. The findings indicate that the generative AI significantly supports users by helping them structure their ideas, enhance narrative coherence, and strengthen their ability to communicate their value proposition effectively. The research positions generative AI as an effective tool for structured, outcome-based learning and competence development in entrepreneurial settings.

Gap between works: While the study effectively validates the role of AI in specific creative tasks within an educational context, it does not explore the complexities of a live, long-term business mentoring relationship. Critically, it does not address the need to ground the creative output in external, verifiable business data (e.g., market reports or financial feasibility studies). The project fills this gap by utilizing the RAG architecture to ensure that the narrative and pitch development support provided by the AI Coach is always informed by accurate and current entrepreneurial knowledge.

Bibliography list MLA Format: George-Reyes, Jorge, et al. "Creating Narrative Scripts Using ChatGPT to Strengthen Entrepreneurial Skills." *Frontiers in Education*, vol. 9, 2024, article 1378564

Source 9: Research Article - Two Educational Science Chatbots as Tools to Support Students' Self-Study

Contribution for the project: This article provides a crucial pedagogical foundation for the AI Coach, directly supporting the project's design philosophy of structured, continuous guidance. It strongly reinforces the findings from the Meta-Analysis (Source 5, Wang et al.) by validating the effectiveness of chatbots in real-world self-study scenarios. The key contribution is demonstrating how conversational systems can be meticulously structured to guide users through complex, non-linear content and foster self-regulation and self-reflection, which are vital skills for successful startup founders.

Major subjects: The major subjects that should be covered by the literature review based on this article include Conversational AI in Education (practical, real-world application of chatbots for learning), Self-Study and Self-Regulation (the ability of AI to support users in managing their own learning process), and Adaptive Tutoring Design (how conversational flow is structured to handle complex knowledge domains).

Summary: The study examines the application of two educational chatbots designed to support university students in self-study within a demanding science course. The research emphasizes the chatbots' capacity to act as structured guides, assisting students in navigating complex material and promoting meta-cognitive skills relevant to learning, namely, self-regulation. Findings show that these structured conversational tools were effective in enhancing the students' learning process and supporting them through independent study phases. This success is directly attributed to the chatbots' ability to maintain a supportive, structured, and goal-oriented dialogue over time.

Gap between works: The article provides an excellent framework for supporting learning, but it is limited to the highly structured environment of academic education. This context differs significantly from the high-stakes, ambiguous, and emotionally charged reality of early-stage entrepreneurship. It also fails to address the unique challenges of integrating dynamic, unstructured business data (as opposed to fixed curriculum knowledge) and mitigating the hallucination risk associated with using LLMs for high-value advice. The project fills this gap by adapting the proven structural and pedagogical effectiveness of these educational chatbots (for self-study) and implementing them within a RAG architecture specialized for the chaotic, evidence-based domain of startup formation.

Bibliography list MLA Format: Neumann, Alexander, et al. "Two Educational Science Chatbots as Tools to Support Students' Self-Study." *Frontiers in Artificial Intelligence*, vol. 4, 2021, article 668220.

Source 10: Working Paper - The Uneven Impact of Generative AI on Entrepreneurial Performance

Contribution for the project: This source is highly valuable because it provides a strong empirical argument for the project's target audience and value proposition. The paper demonstrates that generative AI disproportionately improves the performance of less-skilled or less-experienced entrepreneurs. This justifies the project's focus on early-stage founders who are often defined by their lack of experience and resources, thereby validating the specific target market. Furthermore, it validates that AI reduces the skill gap and cognitive load in entrepreneurial tasks, which directly informs the project's core promise to provide structured guidance where it is most needed. It also suggests that one of the project's success metrics should be the quality of entrepreneurial artifacts (e.g., business plans, pitch ideas) produced with the AI's support.

Major subjects: The major subjects that should be covered by the literature review based on this article include Generative AI and Skill Gaps (how AI narrows performance differences between skilled and less-skilled individuals), AI in Entrepreneurial Tasks (the application of AI for idea generation, planning, and decision support), and Performance Measurement (quantitative evaluation of entrepreneurial output quality under AI influence).

Summary: This Harvard Business School working paper presents findings from a study examining how generative AI affects performance across various entrepreneurial tasks. The key finding is that the impact of the AI is uneven: while highly experienced entrepreneurs see modest gains, novice founders experience the largest and most significant boosts in performance and task quality. The AI functions effectively as a structured support system, helping less-skilled founders overcome "blank-page" problems, structure their thinking, and define objectives clearly. This reduction in the skill gap makes AI a powerful tool for democratizing access to high-quality entrepreneurial planning, reinforcing the project's mission.

Gap between works: The study measures the effect of general, one-time use of LLMs on task completion. It does not examine the efficacy of a long-term, continuously context-aware AI mentor designed to support an entire venture life cycle. Furthermore, the paper measures performance based on output quality but does not delve into the underlying technological architecture (RAG) required to ensure the advice provided is accurate, up-to-date, and grounded in domain-specific knowledge (which is essential for avoiding bias and hallucinations in real-world business advice). The project addresses this by providing the robust, RAG-based technical framework necessary to deliver the consistent, high-impact performance demonstrated to be possible in this study.

Bibliography list MLA Format: Otis, Holmes, et al. *The Uneven Impact of Generative AI on Entrepreneurial Performance*. Working Paper 24-042. Harvard Business School, 2023.

Source 11: Conference Paper - Designing an AI-Powered Mentorship Platform for Personalized Guidance

Name of article: Designing an AI-Powered Mentorship Platform for Personalized Guidance.

Contribution for the project: This conference paper, situated at the intersection of AI and education, offers critical design and architectural guidelines for building the AI Coach. It moves beyond theoretical concepts by detailing the practical requirements for creating a personalized and trustworthy mentorship platform. The core contribution is the strong emphasis on data privacy, user experience (UX) for trust, and a modular architecture that supports individual user needs. This directly informs the project's Non-Functional Requirements (NFRs) in the Project Analysis file, particularly concerning security (NFR5) and usability (NFR4), and validates the decision to use a secure user authentication system (Supabase).

Major subjects: The major subjects that should be covered by the literature review based on this article include AI Mentorship Platform Design (practical design requirements for AI systems that offer long-term guidance), Personalization and Trust in AI Systems (how to design conversational systems that foster user trust and tailor content effectively), and Data Security in Conversational Agents (the necessary security measures required when handling sensitive personal or business data).

Summary: The paper describes the design process and key principles for an AI-powered mentorship platform intended to provide personalized guidance. The authors stress that effective mentorship platforms must not only deliver accurate advice but also be built on a foundation of user trust. This requires rigorous attention to data handling, confidentiality, and transparency in the AI's capabilities and limitations. The platform's design prioritizes a modular structure to adapt guidance based on the user's progress and goals, echoing the need for structured, goal-oriented support seen in Intelligent Tutoring Systems (Source 5). Overall, the work emphasizes that the human-AI interaction in mentorship must be ethically sound and technically reliable to ensure user engagement and long-term utility.

Gap between works: While the paper provides an excellent blueprint for designing a mentorship platform, its focus remains largely on general educational and career guidance, rather than the high-stakes, domain-specific requirements of early-stage entrepreneurial coaching. Crucially, it does not detail the Vector Database or RAG architecture required for indexing large, unstructured business documents (like market reports or financial data) necessary for the Startup Coach. The project

leverages this paper's strong ethical and UX framework while applying a robust RAG architecture to address the unique data challenges inherent in the entrepreneurship domain, thereby filling the technical gap in content customization.

Bibliography list MLA Format: Bagai, Rahul, et al. "Designing an AI-Powered Mentorship Platform for Personalized Guidance." *Proceedings of the 2024 IEEE International Conference on Artificial Intelligence and Education*, 2024, pp. 1–8.

Source 12: Professional Framework - Artificial Intelligence Coaching Framework and Standards: Examples

Name of article: Artificial Intelligence Coaching Framework and Standards: Examples.

Contribution for the project: This document is paramount because it provides the essential ethical and professional grounding for the AI Coach. It moves the project beyond mere technical implementation and into the realm of responsible AI application. The core contribution is establishing the industry's consensus on standards for transparency, data handling, and the boundaries of AI coaching. This directly informs the necessary Prompt Engineering to instruct the AI to clearly state its limitations and distinguish itself from a human coach, as well as the requirement to include metrics that assess the AI Coach's adherence to professional standards and ethical behavior as part of the overall Performance Metrics & Evaluation section. Furthermore, it reinforces the necessity of robust data security and transparency, directly supporting the project's Non-Functional Requirements (NFRs).

Major subjects: The major subjects that should be covered by the literature review based on this source include Ethical AI in Professional Services (defining responsible use and transparency in sensitive advisory roles), AI Coaching Standards and Boundaries (establishing what AI coaches should and should not do according to professional consensus), and Professional Governance of Generative AI (how professional bodies like the ICF adapt to emerging AI technologies and guide their responsible implementation).

Summary: This document outlines the International Coaching Federation's (ICF) framework and standards for integrating artificial intelligence into coaching practices. It provides guidelines for how AI tools should be used responsibly, emphasizing that AI must operate with transparency, clear accountability, and data security. The framework stresses that AI coaches must not overpromise or mislead users, and they should be designed to support rather than replace the human element of reflective and developmental processes. This is a foundational text for anyone building an AI coach, setting the bar for professional compliance and user trust.

Gap between works: As a professional standard and framework, this document is purely theoretical and prescriptive; it dictates *what* should be done but offers zero technical implementation details or empirical data. It does not address the engineering challenges of building the underlying architecture (e.g., RAG, prompt optimization, and contextual retention) required to meet these high standards. The project addresses this by taking the ICF's ethical mandates and translating them into functional and architectural requirements (e.g., using RAG to ensure transparency by citing sources and engineering the prompt to enforce ethical boundaries), thereby bridging the gap between professional standards and computer science implementation.

Bibliography list MLA Format: International Coaching Federation. *Artificial Intelligence Coaching Framework and Standards: Examples*. International Coaching Federation, 2024.

Source 13: Research Article - Comparative Evaluation of an AI-Powered Life Coach against Traditional Coaching Methods

Name of article: Comparative Evaluation of an AI-Powered Life Coach against Traditional Coaching Methods.

Contribution for the project: This is a critical comparative and validation source that directly addresses the core feasibility question of my project: can AI truly substitute or complement human coaching? The study provides empirical evidence showing that AI-based coaching can deliver outcomes comparable to human coaching in certain life domains. This highly validates the central premise of my business coach project that an automated system can achieve developmental results. It also informs the Performance Metrics & Evaluation section by demonstrating how AI coaching efficacy can be quantitatively measured against established human methodologies, such as goal attainment and user satisfaction, offering clear benchmarks for the AI Coach's success criteria. Furthermore, the findings support the argument for market acceptance and user willingness to engage with AI-led guidance.

Major subjects: The major subjects that should be covered by the literature review based on this article include Comparative AI Efficacy (empirical comparisons of AI coaching versus human coaching), Goal Attainment in AI Systems (measuring the effectiveness of AI in driving user-defined goals), and Life Coaching and AI Adaptation (adapting professional coaching methods to conversational AI platforms).

Summary: The article presents a rigorous comparative evaluation of an AI-powered life coach against traditional, human-delivered coaching methods. The study's findings are highly encouraging, demonstrating that the AI system achieved results in goal attainment and user satisfaction that were statistically comparable to those achieved by human coaches in specific coaching areas. This efficacy is attributed to

the AI's structure, consistency, and continuous availability. However, the study also implicitly suggests that, while effective, the AI primarily succeeded in areas requiring structure and less emotional complexity, leaving space for human expertise in nuanced situations.

Gap between works: The study is focused on life coaching, which is a high-structure, less data-intensive domain compared to early-stage entrepreneurial coaching. Entrepreneurship requires accuracy rooted in external, dynamic, and unstructured business data (e.g., market trends, financing rules). This study does not address the technical challenge of grounding the AI's advice in this complex external knowledge base to ensure factual accuracy and avoid hallucination. The project fills this gap by utilizing the robust RAG architecture and specialized Vector Databases to provide the data-grounded, verifiable advice necessary for the high-stakes financial and strategic decision-making in a startup context, which goes beyond the scope of a general life coach.

Bibliography list MLA Format: Terblanche, Nicky, et al. "Comparative Evaluation of an AI-Powered Life Coach against Traditional Coaching Methods." *International Journal of Evidence Based Coaching and Mentoring*, vol. 22, no. 1, 2024, pp. 45-60.

Source 14: Opinion Article

Name of article: 'So What If ChatGPT Wrote It?' Multidisciplinary Perspectives on the Implications of Generative AI for Research, Practice and Policy.

Contribution for the project: This is an extremely valuable source because it provides a broad, multidisciplinary perspective on the strategic and ethical context for deploying any Generative AI application. This reinforces the need for governance, transparency, and accountability in LLM applications (Source 12) as well as a robust Risk Analysis. This study validates how GenAI disrupts professional fields, demonstrating that AI can serve as a consulting tool (Practice) but requires caution with regard to bias and factual accuracy. It elevates the project's academic significance beyond simple technical implementation by placing it within the current global debate on GenAI in human decision-making.

Major subjects: The major subjects that should be covered by the literature review based on this article include Governance and Policy in Generative AI (the regulatory and ethical challenges posed by widespread LLM adoption), Transparency and Accountability (the requirement for AI systems to clearly disclose their origins and reasoning), and Multidisciplinary Impact of LLMs (the strategic implications of AI across various professional sectors).

Summary: This comprehensive article compiles perspectives from 43 experts across multiple disciplines (including IT, management, and policy) regarding the profound

implications of Generative AI like ChatGPT. The authors emphasize that while GenAI offers unprecedented opportunities for efficiency and content creation, it simultaneously introduces critical risks related to intellectual property, data security, algorithmic bias, and academic integrity. The central consensus is the pressing need for researchers and practitioners to develop governance frameworks and strategic policies that ensure responsible, ethical use. It serves as a strong reminder that technology deployment must be accompanied by rigorous evaluation of its ethical and practical consequences on human practice and policy.

Gap between works: In this multidisciplinary opinion and policy piece, the "macro" challenge is defined, but no specific technological solutions or empirical data are provided. Despite identifying the need for transparency and accountability, it does not elaborate on the computer science mechanisms necessary to accomplish them. Using the RAG architecture, the project utilizes the ethical mandate for transparency (by citing retrieved sources) and accountability (by grounding responses in verifiable facts, mitigating bias and hallucination).

Bibliography list MLA Format: Dwivedi, Yogesh K., et al. "'So What If ChatGPT Wrote It?' Multidisciplinary Perspectives on the Implications of Generative AI for Research, Practice and Policy." *International Journal of Information Management*, vol. 71, 2023, article 102642

Source 15: Survey - Retrieval-Augmented Generation for Large Language Models

Name of article: Retrieval-Augmented Generation for Large Language Models: A Survey.

Contribution for the project: This comprehensive survey is a foundational technical source that provides the theoretical framework for my entire RAG architecture and its justification within the context of Large Language Models (LLMs). It validates the project's decision to use RAG by articulating its core function: solving LLM limitations such as hallucination, outdated knowledge, and non-transparent reasoning (Gao et al. 2024). This directly informs the Methodology and Architecture section, detailing the required components like Indexing, Retrieval, and Augmentation.

Major subjects: The major subjects that should be covered by the literature review based on this article include RAG Paradigms (Naive, Advanced, Modular RAG), LLM Limitations (hallucination and knowledge cut-off), and RAG Architecture Components (the functions of the retriever, generator, and augmentation stages).

Summary: The survey examines the rapid evolution of Retrieval-Augmented Generation (RAG) technology, which enhances LLMs by integrating external knowledge bases to improve factual accuracy and credibility. The article systematically categorizes RAG into three developmental paradigms: Naive,

Advanced, and Modular RAG and meticulously analyzes the core technical foundation, including the optimization techniques applied to the retrieval and generation phases. The authors assert that RAG synergistically merges the LLM's intrinsic processing power with vast, dynamic external repositories, making it the key technology for adapting LLMs to real-world, knowledge-intensive applications.

Gap between works: As a general survey on RAG, the article focuses heavily on the technical architecture but does not specify the unique challenges of applying RAG to high-stakes, unstructured business consultation, where the knowledge base must adapt to specific entrepreneurial methodologies. It provides the "how" of RAG but not the "what" (the specific business knowledge) or the "why" (the ethical justification for grounding advice). The project addresses this by taking the robust RAG framework detailed here and applying it to the unique entrepreneurial domain, while adhering to professional ethical standards (Dwivedi et al. 2023; ICF 2024).

Bibliography list MLA Format: Gao, Yunfan, et al. "Retrieval-Augmented Generation for Large Language Models: A Survey." *arXiv*, arXiv:2312.10997, 2024.

Source 16: Survey - When Large Language Models Meet Vector Databases

Name of article: When Large Language Models Meet Vector Databases: A Survey.

Contribution for the project: This source is vital for justifying the selection of Vector Databases as the specific retrieval component in my architecture. It explains why traditional databases are inadequate for RAG and positions Vector Databases as the cost-effective, scalable solution for managing the high-dimensional data (embeddings) required for semantic search.

Major subjects: The major subjects that should be covered by the literature review based on this article include Vector Databases (VDBs) (their necessity and function in the AI ecosystem), Semantic Search and Embeddings (how VDBs enable retrieval based on meaning rather than keywords), and Cost-Effectiveness of RAG (using VDBs to manage knowledge dynamically, avoiding expensive LLM retraining).

Summary: The survey provides a unique analysis of the intersection between Large Language Models (LLMs) and Vector Databases (VDBs). The authors argue that VDBs offer solutions to critical LLM shortcomings, including hallucination, real-time knowledge updates, and high maintenance costs. VDBs excel at handling high-dimensional data, making them crucial for efficient semantic search. By acting as a non-volatile, easily updatable external memory, VDBs allow LLMs to access vast amounts of external, domain-specific information without expensive retraining, thus enhancing overall system accuracy and commercial feasibility.

Gap between works: Like most technical surveys, this article focuses on the "database" aspect and does not address the user experience, ethical governance, or

pedagogical framework (ITS principles) necessary for turning the technology into a successful coaching product. It provides the technical 'engine,' but not the 'guidance system.' The project fills this gap by integrating the Vector Database technology detailed here with the coaching frameworks (ICF, Terblanche) and the conversational design principles (Akpan, Mariani) to deliver a holistic, productized solution.

Bibliography list MLA Format: Jing, Zhi, et al. "When Large Language Models Meet Vector Databases: A Survey." *arXiv*, arXiv:2402.01763, 2024.

Source 17: Survey - A Survey on RAG Meeting LLMs

Name of article: A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models.

Contribution for the project: This detailed technical survey is essential for exploring the advanced customization and training options within the RAG framework. It reinforces the need for RAG by explicitly detailing LLM limitations and provides the necessary detail for the "Future Work" section in my Project Analysis by discussing concepts like Iterative Retrieval and Augmentation Strategies (Fan et al. 2024).

Major subjects: The major subjects that should be covered by the literature review based on this article include RAG Training Strategies, Retrieval Augmentation Integration, and Advanced RAG Challenges (trustworthiness and robustness).

Summary: This comprehensive survey provides a systematic review of Retrieval-Augmented Large Language Models (RA-LLMs), covering their architectures, training strategies, and applications. The authors highlight RAG's ability to provide reliable, up-to-date external knowledge, thereby mitigating issues like hallucination and out-of-date internal knowledge in LLMs. The survey provides detailed diagrams and descriptions of the RAG components, focusing on how retrieved information is effectively integrated into the generation process. It concludes by discussing current limitations, such as the need for more trustworthy and robust RAG systems, which directly informs the Risk Analysis and Future Work sections of my project.

Gap between works: This is a purely technical survey focused on algorithmic mechanisms (training, integration layers). It lacks context on the domain-specific content (entrepreneurship), human factors (trust, empathy, coaching ethics), and the user-centric metrics required to validate a non-technical product like a business coach. The project bridges this by marrying the detailed technical strategies presented here with the ICF standards (Source 12) and performance metrics (Source 13), ensuring the technical complexity serves a validated, ethical, and usable outcome.

Bibliography list MLA Format: Fan, Wenqi, et al. "A Survey on RAG Meeting LLMs: Towards Retrieval-Augmented Large Language Models." *Proceedings of the 30th*

Project Analysis - 2 חלק

Project Competitor Analysis: StartupCoach

Competitor Analysis: Existing AI Advisory Tools

1. ChatGPT Enterprise / Custom GPTs (OpenAI)
 - Primary Target Audience: General users, professionals, enterprises, and developers across diverse domains
 - Business Model: Subscription-based (Enterprise licensing, API usage).
 - Strengths: Unrivaled LLM general knowledge base; excellent natural language fluency and coherence. Easy to prompt for creating basic personas (e.g., "Act as a coach").
 - Weaknesses: Susceptible to hallucination and knowledge cutoff (not up-to-date); lacks continuous, personal context storage; cannot be reliably grounded in specific, proprietary business documents (requires RAG).
 - Differentiator: ChatGPT sets the benchmark for generative fluency and general-purpose reasoning. StartupCoach does not compete on raw language generation, but on process awareness, domain grounding, and decision support for a specific entrepreneurial niche.
2. Jasper / Copy.ai (Generative Content Platforms)
 - Primary Target Audience: Marketing teams, content creators, and growth-focused organizations.
 - Business Model: Subscription SaaS for content generation.
 - Strengths: Excellent at structured content generation (marketing copy, outlines, emails); built-in templates streamline routine tasks. Highly user-friendly.
 - Weaknesses: Primarily focused on marketing and sales content, not on strategic or financial advice. Has no capability for conversational, adaptive mentorship or long-term context retention.
 - Differentiator: These platforms excel at *how to say something*, not *what to do next*. StartupCoach targets founders before marketing execution, when strategic uncertainty is highest.

3. CoachVox AI (Industry AI Coaching Tool)

- Primary Target Audience: Professional coaches, consultants, and content creators seeking to scale their existing expertise.
- Business Model: B2B SaaS for creating branded AI coaches.
- Strengths: Directly targets the AI coaching market; high focus on prompt engineering and persona stability.
- Weaknesses: Relies heavily on prompting only, meaning quality is inconsistent, and the advice is not grounded in verifiable external data, making it prone to generic outputs. Lacks transparency in the source of advice.
- Differentiator: CoachVox enables “AI versions of human coaches.” StartupCoach instead aims to embed a coherent entrepreneurial methodology, supported by RAG, rather than replicating individual coaching styles.

4. Intelligent Tutoring Systems (ITS) Conceptual Competitor (e.g., Coursera / Udacity)

- Primary Target Audience: Students and learners in well-defined academic domains.
- Business Model: Course-based or subscription learning platforms.
- Strengths: Proven academic framework for delivering structured, adaptive learning and testing. Effective in improving outcomes in closed domains (like coding).
- Weaknesses: Designed for academic content (Computer Science concepts), not the open, uncertain, and unstructured domain of entrepreneurship. Lacks the Generative AI fluency required for natural conversation.
- Differentiator: ITS platforms provide valuable educational structure, but lack the flexibility and generative reasoning required for entrepreneurial coaching. StartupCoach adapts ITS principles to an LLM-driven, real-world decision context.

5. Human Consultants / Executive Coaches (Traditional Competitor)

- Primary Target Audience: Founders, executives, and organizations seeking personalized guidance.
- Business Model: High-cost, time-based professional services.
- Strengths: High emotional intelligence and contextual sensitivity. Ability to handle nuanced, high-stakes situations. Access to personal networks and tacit knowledge (Jarco et al., 2023).

- Weaknesses: High cost, low availability (not 24/7), and limited scalability (cannot serve many startups simultaneously). Prone to human bias and subjective advice.
- Differentiator: Human coaches represent the gold standard in trust and relational depth. StartupCoach aims to approximate some of these qualities through persona stability, ethical constraints, and consistent methodology, while offering 24/7 availability.

6. Corporate AI Coaching Platforms (e.g., BetterUp, CoachHub, Cloverleaf)

- Primary Target Audience: Enterprises, HR departments, and leadership development programs.
- Business Model: Enterprise SaaS with organizational contracts.
- Strengths: Integrate behavioral science, coaching frameworks, and organizational data. Strong focus on leadership development and performance improvement. Often embedded into workplace tools (Slack, Teams).
- Weaknesses: Designed for employees and managers, not solo founders. High cost and operational complexity. Require organizational infrastructure and buy-in.
- Differentiator: These platforms address corporate coaching at scale, whereas StartupCoach focuses on the individual early-stage founder, operating alone or with a small team.

Strategic Requirements and Design Decisions

Across all analyzed solutions, several common gaps emerge:

Lack of stateful, longitudinal understanding of individual entrepreneurs. Absence of a consistent, auditable coaching methodology grounded in recognized entrepreneurial frameworks. Minimal emphasis on measuring decision quality or learning outcomes, rather than engagement alone.

StartupCoach addresses this gap by combining LLM capabilities, Retrieval-Augmented Generation, tool-enabled reasoning, and structured coaching flows, resulting in a founder-first AI coaching system tailored to early-stage entrepreneurial decision-making.

Natural Language Chat Interface (must have): The system allows the user to enter questions in natural language and receive textual responses from the LLM. It retains the current conversation history so that the bot can remember information provided in previous messages, such as investment amounts stated earlier. Includes functionality for the user to explicitly clear the conversation history and start a new session.

Intent Recognition and Parameter Extraction (must have): The model automatically detects when the user is requesting a business success evaluation and invokes the appropriate tool. It extracts specific parameters required for the ML model (industry, investment amount, years of experience) directly from the user's free text. If the user has not provided all required parameters, the bot identifies the gap and asks follow-up questions to obtain the missing data before invoking the tool.

ML-Based Prediction Engine (must have): A dedicated Python server handles structured input by receiving JSON requests containing the data collected from the chat. It executes a trained ML model (sourced from Kaggle) on the received data to perform the analysis. The server calculates and returns the business success probability (as a decimal value) back to the language model for presentation to the user.

Non-Functional Requirements

System Responsiveness and Latency (must have): Ensures a standard chat response is returned in less than 3 seconds. When the prediction tool is invoked, the full response loop (including network communication with the Python server) completes in less than 5 seconds.

Usability and Persona (must have): The LLM consistently maintains the tone of an empathetic and professional business coach throughout the entire conversation. The user interface is responsive and displays correctly on both desktop and mobile devices to support accessible usage.

Reliability and Security (must have): Manages API keys for OpenAI/Gemini securely by storing them as server-side environment variables, ensuring no keys are exposed in client-side code. Includes graceful error handling mechanisms; if the ML server is unavailable, the bot responds with a user-friendly message instead of crashing.

Future / Optional Features (Nice to Have)

User Authentication and Persistence (optional): A login system allows users to save their conversation history and prediction results permanently across different sessions and devices. This feature is not critical for the core prediction logic but enhances long-term usability.

Advanced Visualization Dashboard (optional): Integrating visual graphs (e.g., Radar Charts, Market Comparison Bars) directly into the chat interface to visually represent the user's strengths and weaknesses relative to the market average.

Real-Time Data Integration (optional): Connecting the backend to live APIs (such as Crunchbase API) to fetch real-time market data, replacing the static CSV-based model training with live updates.

Export to PDF (optional): A feature allowing the user to download a summary of the business analysis and the conversation log as a formatted PDF file for external use.

Architectural Requirements - 4 חלק

High-Level Component Interaction: The system is built on a Hybrid Client-Server Architecture. The Client (Next.js) handles the user interface and chat management. It communicates with two distinct backends: The AI Provider (OpenAI/Gemini) for text generation via the Vercel AI SDK, and a dedicated Python Microservice for deterministic business predictions based on the trained ML model. Supabase manages data persistence.

Prediction ML Model (must have): Scikit-learn-based Random Forest Classifier. Trained on a balanced subset of the Crunchbase dataset (approx. 6,000-10,000 records). Architecture includes an ensemble of decision trees optimized for tabular data. The model achieves over 75% validation accuracy on historical data. Input format is a structured JSON vector (Funding, Milestones, Relationships, Category). Output is a binary classification probability (0-1) representing success potential. Model serialized as a .joblib file for fast inference.

Prediction API Microservice (must have): FastAPI (Python) framework for serving the ML model. Exposes a RESTful endpoint (/predict) to receive JSON data from the frontend. Includes input validation using Pydantic schemas to ensure data integrity (e.g., non-negative funding amounts). Implements Data Normalization logic to adjust historical financial figures for inflation before feeding them to the model. Handles CORS to allow secure communication with the Next.js frontend.

Client-Side Web Application (must have): Next.js 14 (App Router) framework using React components. Implements a Responsive Web App (RWA) design using Tailwind

CSS to ensure a native-app-like experience on mobile devices. Utilizes Vercel AI SDK to manage the chat stream and handle "Function Calling" (routing between text responses and the Python prediction tool). Integrates the Recharts library for rendering dynamic visualizations (Radar Charts, Bar Graphs) directly in the chat interface.

Data Persistence Layer (must have): Supabase (PostgreSQL) database for storing chat sessions and user interaction logs. Utilizes Row Level Security (RLS) policies to ensure data privacy. Stores structured logs of prediction requests and results for future model fine-tuning.

Core Libraries & Environment (must have): Python 3.9+ runtime for the backend. Pandas and NumPy for data pre-processing and vectorization. Scikit-learn for model execution. Uvicorn as the ASGI web server implementation. Development and training conducted in Google Colab / Jupyter Notebooks, with final deployment to a cloud environment (e.g., Render/Vercel).

Technologies Planned for Future/Optional Implementation

Production Infrastructure: Docker containerization for the Python microservice to ensure environment consistency. Redis caching to store frequent prediction results and reduce model inference load.

Enhanced Data Pipeline: Integration with Crunchbase Live API to replace the static CSV dataset with real-time market data updates. Implementation of an automated retraining pipeline (MLOps) that updates the model monthly based on new industry trends.

Advanced User Features: Supabase Auth for secure user login and profile management. PDF export generation using React-PDF to allow users to download a formal business assessment report. Dark/Light mode theming.

Development Environment (must have) IDE: Visual Studio Code (latest version) with extensions for Python, React/Next.js, and Tailwind CSS IntelliSense.

Version Control: Git for source code management, with a remote repository hosted on GitHub.

Runtime Environments: Node.js (LTS version 20+) for frontend execution and Python 3.9+ for the backend microservice.

Hardware Requirements (must have) Development Machine: Standard PC/Laptop with at least 8GB RAM (16GB recommended) and internet connection for API access.

Server/Cloud: The system requires a cloud hosting environment (e.g., Vercel for Frontend, Render/Railway for Backend) capable of running Docker containers or Python processes.

Software Interfaces & APIs (must have) LLM Provider: OpenAI API (GPT-4o/Turbo) or Google Gemini API via Vercel AI SDK. Requires a valid API Key.

Database Connection: Supabase Client (@supabase/supabase-js and supabase-py) connecting via HTTPS/WebSockets.

Data Source: Integration with the Crunchbase CSV dataset (processed via Pandas) locally or stored in cloud storage.

External Libraries and Frameworks (must have) Frontend: Next.js 14, React 18, Tailwind CSS, Lucide React (icons), Recharts (graphs).

Backend: FastAPI, Uvicorn, Pydantic.

Data Science: Scikit-learn, Pandas, NumPy, Joblib (for model serialization).

חלק 6 – Detailed design

System Architecture ((ארכיטקטורת המערכת))

The system is designed using a Hybrid Microservices Architecture. It decouples the User Interface (Client) from the computationally intensive Logic (ML) and the Generative capabilities (LLM).

The 4 Main Components:

Frontend Client (Next.js): Manages user interaction, state, and visualization. Acts as the orchestrator.

AI Orchestration Layer (Vercel AI SDK): Manages the context window and streams responses from OpenAI/Gemini.

Prediction Microservice (Python FastAPI): A dedicated stateless service that runs the Random Forest model for business success prediction.

Persistence Layer (Supabase/PostgreSQL): Stores relational data (users, history, logs).

Modular Decomposition ((פירוק למודולים))

A. Frontend Module (Next.js)

ChatComponent: Handles message input/output and streaming.

GraphVisualizer: Renders Radar/Bar charts using Recharts.

AuthHandler: Manages user sessions via Supabase Auth.

B. Backend Prediction Service (Python)

DataPreprocessor: Normalizes input currency and scales values.

ModelLoader: Loads the .joblib model file into memory on startup.

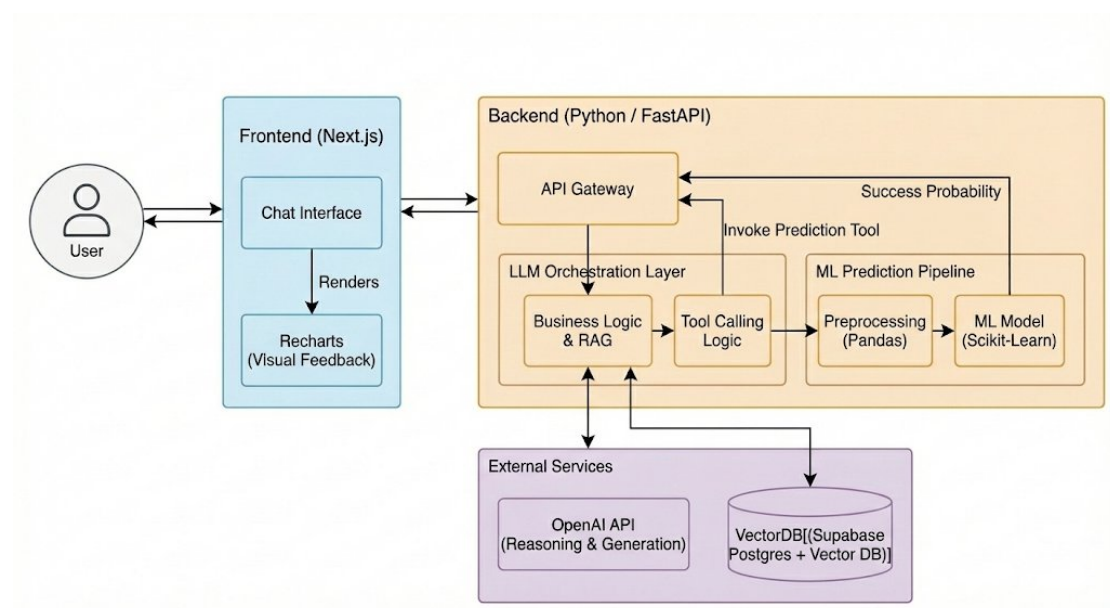
InferenceEngine: Runs the prediction logic (model.predict_proba).

C. Database Module (Supabase)

PostgresDB: The relational storage engine.

VectorStore: For semantic search capabilities.

Diagram ((דיאגרמה))



Technical Tradeoffs ((השוואות טכנולוגיות))

Tradeoff 1: Database Selection (PostgreSQL vs. MongoDB)

- Option A: MongoDB (NoSQL)

Pros: Flexible schema, fast setup.

Cons: No strict data integrity, weak relationships.

- Option B: PostgreSQL/Supabase (Selected)

Pros: Strong data integrity (ACID), handles relationships (Users->Chats) efficiently, built-in Vector support (pgvector).

Cons: Schema migration required for changes.

- **Decision:** Selected PostgreSQL because financial data requires a strict structure, and relationships are critical.

Tradeoff 2: ML Hosting (Python Microservice vs. Node.js ONNX)

- **Option A: Run Model in Node.js (Client-side/Serverless)**

Pros: Simpler architecture (no extra server).

Cons: Limited ML libraries in JS, slower processing for heavy math.

- **Option B: Python Microservice (Selected)**

Pros: Access to full Scikit-learn/Pandas ecosystem, separate scaling for ML logic.

Cons: Adds latency (network call), requires managing a separate container.

- **Decision:** Selected Python Microservice to utilize the robust Pandas/Scikit-learn libraries required for accurate data processing.

7. Pseudocode ((פסאודו-קוד ללוגיקה המרכזית))

```
9 FUNCTION handle_user_request(user_input):
10     # Step 1: Analyze intent using LLM
11     intent = LLM.classify(user_input)
12
13     IF intent IS "business_evaluation":
14         # Step 2: Extract structured data
15         params = LLM.extract_entities(user_input)
16
17         # Step 3: Validate Data
18         IF params.funding IS missing OR params.category IS missing:
19             RETURN "Please provide funding amount and category."
20
21         # Step 4: Call Prediction Service
22         normalized_data = preprocess(params)
23         probability = ML_Model.predict(normalized_data)
24
25         # Step 5: Save to DB
26         Database.log_prediction(params, probability)
27
28         RETURN format_response(probability)
29
30     ELSE:
31         # Standard chat flow
32         RETURN LLM.generate_response(user_input)
```